## 1.11 Inequalities, Convergence, and Limit Theorems of RVs

### 1.11.1 Inequalities

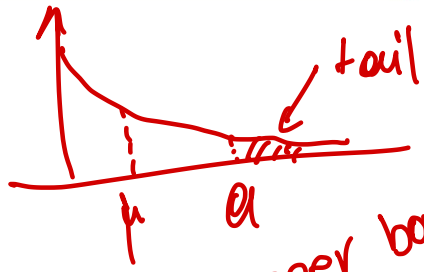
**Markov's inequality**

*If a random variable X takes nonnegative values, then for any $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{E(X)}{a}.$$

proof

Let $Y = \begin{cases} a & \text{if } X \geq a \\ 0 & \text{if } X < a \end{cases}$




upper bound of tail probability

Because X is nonnegative

$X \geq Y \Rightarrow E(X) \geq E(Y) = 0\,P(X<a) + a\,P(X \geq a)$

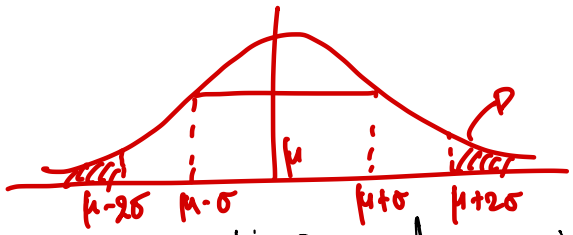
$\Rightarrow E(X) \geq a\,P(X \geq a)$

**Chebyshev's inequality**

*If X is a random variable with mean $\mu$ and variance $\sigma^2$, then for any $k > 0$,*

$$\mathbb{P}\left(|X - \mu| \geq k\sigma\right) \leq \frac{1}{k^2}.$$

proof




$\frac{(X-\mu)^2}{\sigma^2}$ is a nonnegative random variable

and its mean is

$$E\left(\frac{(X-\mu)^2}{\sigma^2}\right) = \frac{\overbrace{E[(X-\mu)^2]}^{\to Var(X)}}{\sigma^2} = 1$$

gives a bound on the prob that X deviates significantly from its mean

Using Markov's inequality, we get

$$P\left[\frac{(X-\mu)^2}{\sigma^2} \geq k^2\right] \leq \frac{E\left[\frac{(X-\mu)^2}{\sigma^2}\right]}{k^2} = \frac{1}{k^2}.$$

**Jensen's Inequality**

*Let $\phi : \mathbb{R} \to \mathbb{R}$ be a convex function, and let $X$ be a random variable. Then,:*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Proof

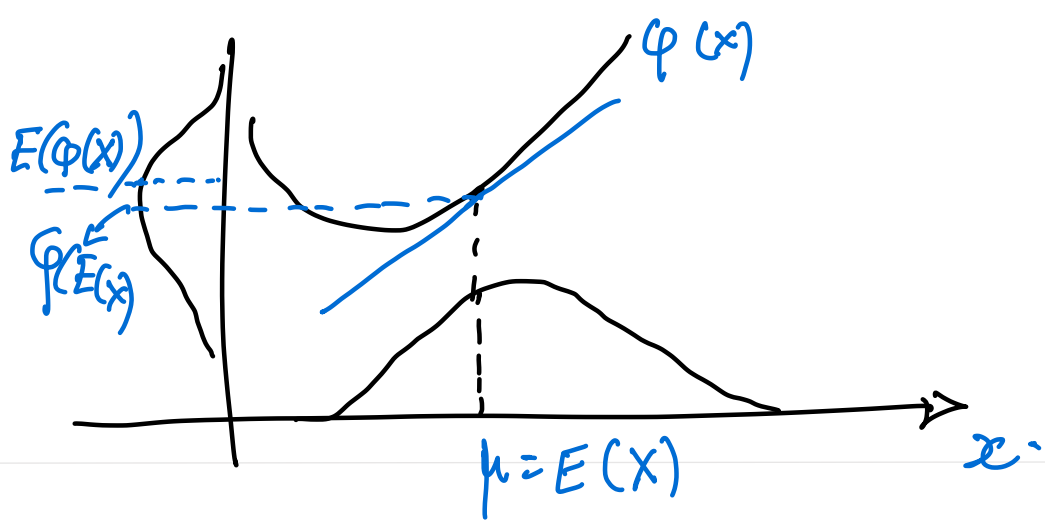
Let $\mu = E(x)$. Since $\varphi$ is convex, $\phi(x)$ is above the tangent at $\mu$.

$$\varphi(x) \geq \varphi(\mu) + \varphi'(\mu)(x-\mu)$$

By taking the expectation, we have

$$E(\varphi(X)) \geq E(\varphi(\mu)) + \varphi'(\mu)\cancelto{0}{(E(x)-\mu)}$$

$$\rightarrow \boxed{E(\varphi(X)) \geq \varphi(E(X))}$$



### 1.11.2 Convergence of Sequence of Random Variables

**Definitions**

*Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Consider a random variable $X$ and a sequence of random variables $X_1, X_2, \ldots$ all defined on the same probability space. Then*

**Almost sure convergence**

*The sequence converges to $X$ **almost surely**, shown by $X_n \xrightarrow{a.s.} X$, if*

$$\mathbb{P}\left(\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right) = 1.$$

<u>Example</u> Consider $X_n(\omega) = \omega^n$, $n = 1, 2, \ldots$ and $\omega \in [0,1] = \underline{\Omega}$. Show that $X_n \xrightarrow{a.s.} X = 0$

Let $A = \{\omega \in \underline{\Omega} : \lim_{n\to\infty} X_n(\omega) = 0\}$

we need to show that $\mathbb{P}(A) = 1$

Note that for all $\omega \in [0,1)$, $\lim_{n\to\infty} \omega^n = 0$

Only $\omega = 1$ gives $\lim_{n\to\infty} \omega^n = 1 \neq 0$, $\forall n$.

Thus $A = \underline{\Omega} - \{1\}$, but since $\mathbb{P}(\{1\}) = 0$

$$\mathbb{P}(A) = \mathbb{P}(\underline{\Omega}) - \mathbb{P}\{1\}$$

$$= 1 \qquad \text{So} \qquad X_n \xrightarrow{a.s.} 0.$$

If $A = \underline{\Omega}$, we have <u>sure</u> convergence

**Convergence in probability**

*The sequence* ***converges in probability*** *to $X$, shown by $X_n \xrightarrow{p} X$, if for any $\epsilon > 0$,*

$$\lim_{n\to\infty} P\left(|X_n - X| < \epsilon\right) = 1.$$

$> \varepsilon = 0$

<u>Example</u> Let $X_n = \begin{cases} n & 1/n \\ 0 & 1-1/n \end{cases}$

Show that $X_n \xrightarrow{P} X = 0$

$$\lim_{n\to\infty} P(|X_n - 0| > \varepsilon) = \lim_{n\to\infty} P(|X_n| > \varepsilon)$$

$$= \lim_{n\to\infty} P(X_n = n) = \lim_{n\to\infty} P\left(\frac{1}{n}\right) = 0$$

$\forall\, \varepsilon > 0$

**Example 1.10** Let

$$X = \begin{cases} 0, & \text{with probability } \frac{1}{2} \\ 1, & \text{with probability } \frac{1}{2} \end{cases}.$$

Then $X_n = (1 + 1/n)X$ converges to $X$ in probability. Note that

$$X_n - X = \left(1 + \frac{1}{n}\right) X - X = \frac{1}{n} X.$$

Then for $\epsilon > 0$, if $X = 0$, we have

$$|X_n - X| = \frac{1}{n} \cdot 0 = 0 \quad \Longrightarrow \quad \lim_{n\to\infty} P\left(|X_n - X| > \epsilon\right) = 0.$$

If $X = 1$, we have

$$|X_n - X| = \frac{1}{n} \cdot 1 = \frac{1}{n} \quad \Longrightarrow \quad \lim_{n\to\infty} P\left(|X_n - X| > \epsilon\right) = 0.$$

**Convergence in mean square**

*The sequence* ***converges in mean square*** *to* $X$*, shown by* $X_n \xrightarrow{m.s.} X$*, if*

$$\lim_{n\to\infty} E\left((X_n - X)^2\right) = 0.$$

**Example 1.12** Consider a random variable $Y \sim \mathcal{U}([0,1])$ and a sequence of random variables $X_n(y)$ such that

$$X_n = \begin{cases} a_n, & \text{if } y < \frac{1}{n} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$E\left((X_n - 0)^2\right) = a_n^2 P(X_n = a_n) + 0^2 P(X_n = 0) = a_n^2 \frac{1}{n}.$$

Now consider the choices of $a_n$,

1. If $a_n = 1$, then $X_n \xrightarrow{\text{m.s.}} 0$,
2. If $a_n = \sqrt{n}$, then
$$\lim_{n\to\infty} E\left((X_n - 0)^2\right) = \lim_{n\to\infty} n \cdot \frac{1}{n} = 1,$$
3. If $a_n = n^2$, then
$$\lim_{n\to\infty} E\left((X_n - 0)^2\right) = \lim_{n\to\infty} n^2 \cdot \frac{1}{n} = n.$$

**Convergence in distribution**

*The sequence* ***converges in distribution*** *to $X$, shown by $X_n \xrightarrow{d} X$, if*

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

*for all $x$ at which $F_X(x)$ is continuous.*

**Example 1.13** Let $X_1, X_2, \ldots$ be a sequence of random variables with CDF

$$F_{X_n} = 1 - \left(1 - \frac{1}{n}\right)^{nx}.$$

Then $F_{X_n}$ converges to exponential distribution. Note that

$$\begin{aligned}
\lim_{n\to\infty} F_{X_n} &= \lim_{n\to\infty} \left[1 - \left(1 - \frac{1}{n}\right)^{nx}\right] \\
&= 1 - \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^{nx} \\
&= 1 - e^{-x} \\
&= F_X(x)
\end{aligned}$$

where $F_X(x)$ is the exponential CDF.

**Remark** Note that

1. sLLN is an example of almost surely convergence,
2. wLLN is an example of convergence in probability,

## Proposition

if $E(|X_n|^2) < \infty$ and

if $\sum_{n=1}^{\infty} E[|X_n|^2] < \infty$

then $X_n \xrightarrow{a.s} 0$.

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Proof: Use Chebychev's inequality

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Borel-Cantelli lemma

# Connection between different convergence modes

$$\begin{tikzcd}
a.s \arrow[dr] & & m.s \arrow[dl] \\
& P \arrow[ul, red, "\times"] \arrow[ur, red, "\times"] \arrow[d] & \\
& d \arrow[u, red, "\times"] &
\end{tikzcd}$$

① $X_n \xrightarrow{m.s.} X \Rightarrow X_n \xrightarrow{P} X$

Convergence in $P$ is:

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = \lim_{n\to\infty} P(|X_n - X|^2 > \varepsilon^2)$$

$$\leq \lim_{n\to\infty} \frac{E[|X_n - X|^2]}{\varepsilon^2} = 0$$

Markov's inequality

m.s.

So,

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$$

(2) $X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X.$

For $\varepsilon > 0$.

$$F_{X_n}(x) = P(X_n \leq x)$$

$x < X - \varepsilon$

$$= P(X_n \leq x, X \leq x+\varepsilon) + P(X_n \leq x, X > x+\varepsilon)$$

$$\leq P(X \leq x+\varepsilon) + P(X_n \leq X - \varepsilon)$$

$$= F_X(x+\varepsilon) + P(X_n - X \leq -\varepsilon)$$

$$\leq F_X(x+\varepsilon) + P(|X_n - X| > \varepsilon) \quad (A)$$

Similarly

$$F_X(x-\varepsilon) \leq F_{X_n}(x) + P(|X_n - X| > \varepsilon) \quad (B)$$

Combine A and B to get

$$F_X(x-\varepsilon) - P(|X_n - X|) \leq F_{X_n}(x) \leq F_X(x+\varepsilon) + P(|X_n - X| > \varepsilon)$$

Taking the limit $n \to \infty$, we have

$$F_X(x-\varepsilon) \leq \liminf_{n\to\infty} F_{X_n}(x) \leq \limsup F_n(x) \leq F_X(x+\varepsilon)$$

For $F_X$ continuous at $x$ and since the ineq. holds for any $\varepsilon > 0$, we have

$$\lim F_{X_n}(x) = F_X(x)$$

### 1.11.3 Limit Theorems

**The weak Law of Large Numbers (wLLN)**

*Let $X_1, X_2, \ldots, X_n$ be a sequence of iid random variables with mean $\mu$. Then for any $\epsilon > 0$,*

$$\mathbb{P}\left(|\overline{X}_n - \mu| > \epsilon\right) \to 0$$

*as $n \to \infty$, where*

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

*is the arithmetic mean.*

**Proof** Note that $\mu = E(X_i)$ is the true mean. The arithmetic mean $\overline{X}_n$ is another random variable such that

$$E\left(\overline{X}_n\right) = \frac{E(X_1) + E(X_2) + \cdots + E(X_n)}{n} = \frac{n\mu}{n} = \mu,$$

and

$$Var\left(\overline{X}_n\right) = \frac{Var(X_1) + Var(X_2) + \cdots + Var(X_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. = \Sigma$$

Thus, by the Chebyshev's inequality, we have

$$\mathbb{P}\left(|X_n - \mu| \geq k\cancel{\sigma}\Sigma\right) = \mathbb{P}\left(|X_n - \mu| \geq k\frac{\sigma}{\sqrt{n}}\right) \leq \frac{1}{k^2}$$

which implies that

$$\mathbb{P}\left(|X_n - \mu| \geq \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2} \quad \to 0$$

as $n \to \infty$.

**Central Limit Theorem (CLT)**

*Let $X_1, X_2, \ldots, X_n$ be a sequence of iid random variables having mean $\mu$ and variance $\sigma^2$. Then*

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{\overline{X}-\mu}{\frac{\sigma}{\sqrt{n}}} < x\right) = \Phi(x),$$

*or*

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n}\,\sigma} < x\right) = \Phi(x)$$

*where $S_n = X_1 + X_2 + \cdots + X_n = n\overline{X}_n$ and $\Phi(x)$ is the standard normal distribution.*

**Remark** Note that for the first equation

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{\overline{X}-\mu}{\frac{\sigma}{\sqrt{n}}} < x\right) = \Phi(x),$$

the random variable

$$Z = \frac{\overline{X}-\mu}{\sigma} = \frac{\overline{X}-\mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\overline{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim \mathcal{N}(0,1),$$

and so as $n \to \infty$, we get

$$\mathbb{P}(Z < x) = \Phi(x).$$

From the above, we can write

$$\overline{X} = \mu + \frac{\sigma}{\sqrt{n}} Z,$$

that is, $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ and thus, $S_n \sim \mathcal{N}(n\mu, n\sigma^2)$. Therefore, we get the second equation

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n}\,\sigma} < x\right) = \Phi(x).$$